

Forecasting of COVID-19 per regions using ARIMA models and polynomial functions

Andres Hernandez-Matamoros, Hamido Fujita, Toshitaka Hayashi, Hector Perez-Meana
2020

Morgunov Iliya
Alfasi Yagel
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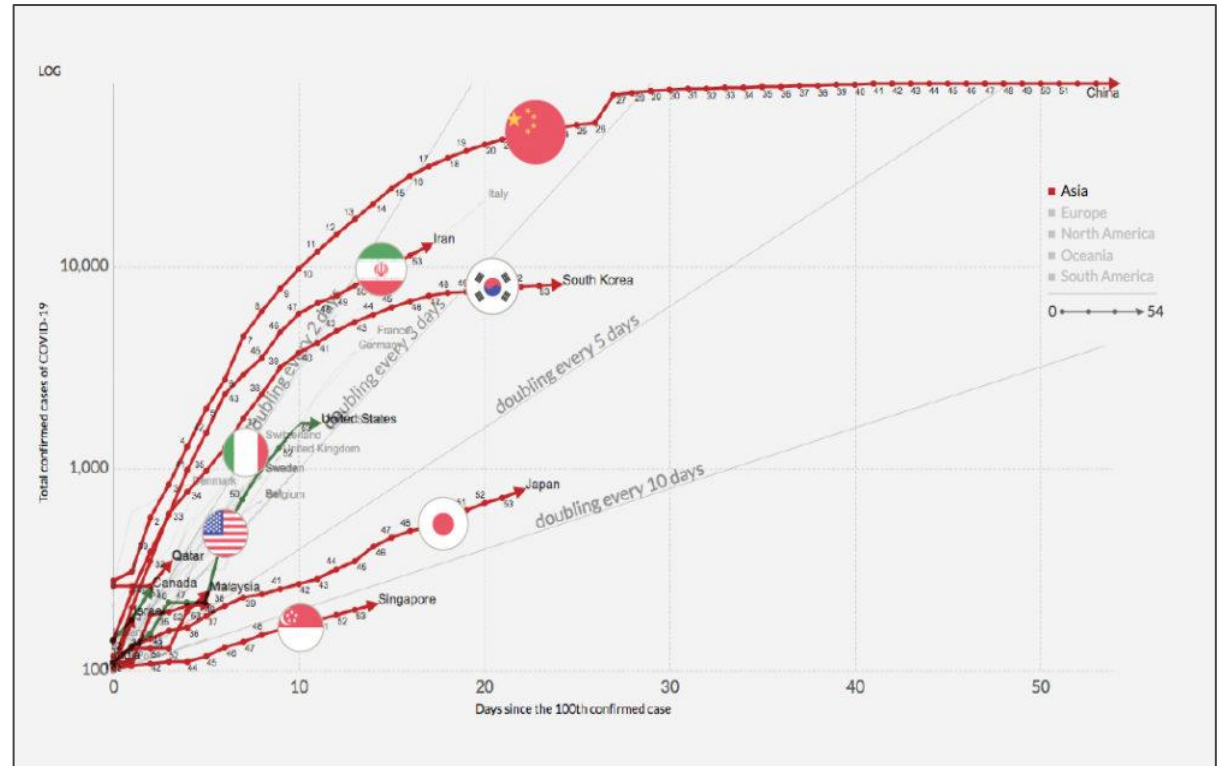


Background

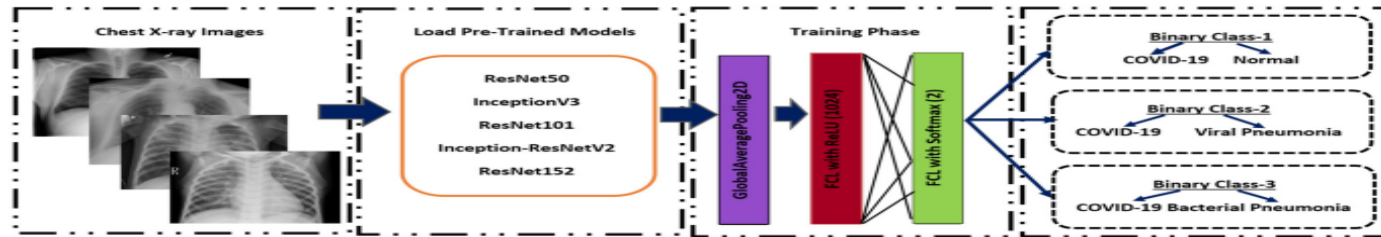
6,750,521
active cases

395,779
deaths

216
Countries



Back to Mid 2020 – What did the researchers do?



Ali Narin, Ceren Kaya, Ziynet Pamuk

More than 96% accuracy!

Asymptomatic novel coronavirus pneumonia patient outside Wuhan, Chen Lin



Prediction

other research groups at the time

Domenico Benvenuto, Marta Giovanetti, Lazzaro Vassallo, Silvia Angeletti, Massimo Ciccozzi, Application of the ARIMA model on the COVID-2019 epidemic dataset, in: Data in Brief, Vol. 29, 2020, 105340

Fong Simon, Gloria Li, Nilanjan Dey, Ruben Gonzalez Crespo, Enrique Herrera-Viedma, Finding an accurate early forecasting model from small dataset: A case of 2019-nCoV novel coronavirus outbreak, Int. J. Interact. Multimed. Artif. Intell. 6 (2020)

Gaetano Perone, An ARIMA model to forecast the spread and the final size of COVID-2019 epidemic in Italy, 2020

Guorong Ding, Xinru Li, Yang Shen, Brief Analysis of the ARIMA model on the COVID-19 in Italy, medRxiv 2020.04.08.20058636

Previous Models

- Were not accurate enough
- Too specific – for e certain country, age range, etc

Main problem

How to generalize?

The purposed method – Objectives

Prediction

Fast

Accurate

General

How?

1. Divide countries into continents
2. Predict for each country each continent
3. Build the model for the continent
4. Make prediction for the country

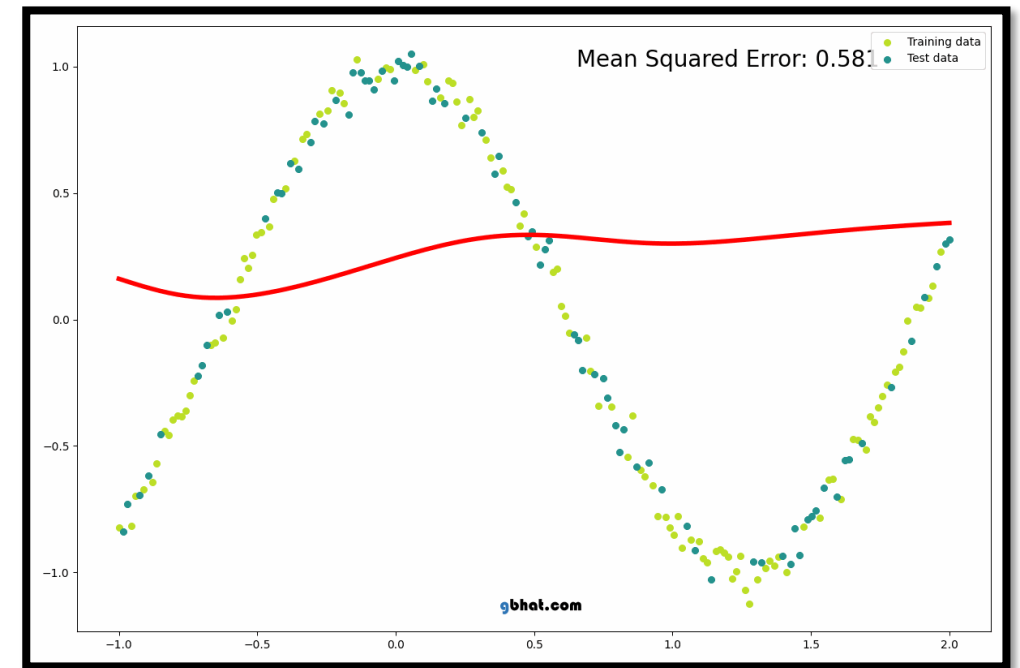
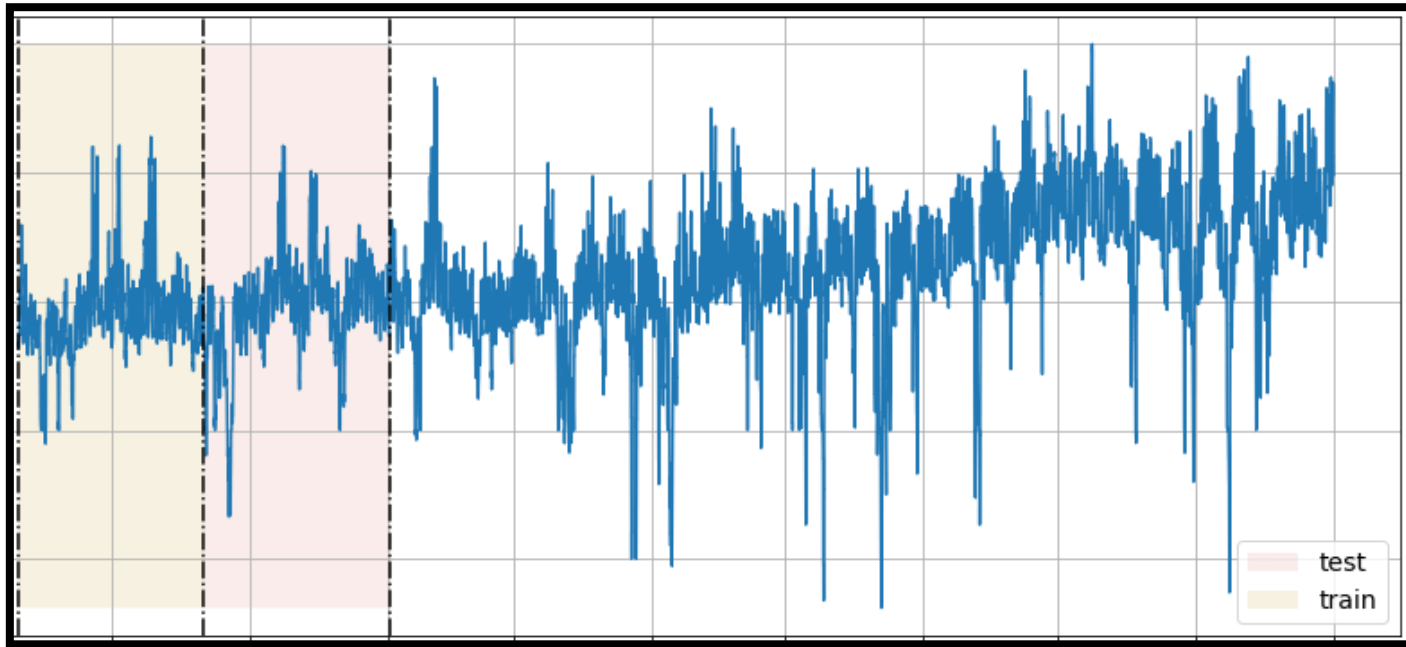


Why?

Maximize the generalization.

Countries in the same geographic region have variables with similar values (quantifiable and non-quantifiable), which affect the spread of the virus.

Methods



Coronavirus Pandemic (COVID-19)

Research and data: Edouard Mathieu, Hannah Ritchie, Lucas Rod s-Guirao, Cameron Appel, Daniel Gavrilov, Charlie Giattino, Joe Hasell, Bobbie Macdonald, Saloni Dattani, Diana Beltekian, Esteban Ortiz-Ospina, and Max Roser

🔄 The data on the coronavirus pandemic is updated daily. Last update: 7 hours ago. © [Reuse our work freely](#) 📄 [Cite this research](#)

Coronavirus > [By country](#) [Data explorer](#) [Deaths](#) [Cases](#) [Tests](#) [Hospitalizations](#) [Vaccinations](#) [Mortality risk](#) [Excess mortality](#) [Policy responses](#)



Explore all metrics – including cases, deaths, testing, and vaccinations – in one place.



Get an overview of the pandemic for any country on a single page.



Download our complete dataset of COVID-19 metrics on GitHub. It's open access and free for anyone to use.



Explore our global dataset on COVID-19 vaccinations.



See state-by-state data on vaccinations in



Explore the data on confirmed COVID-19



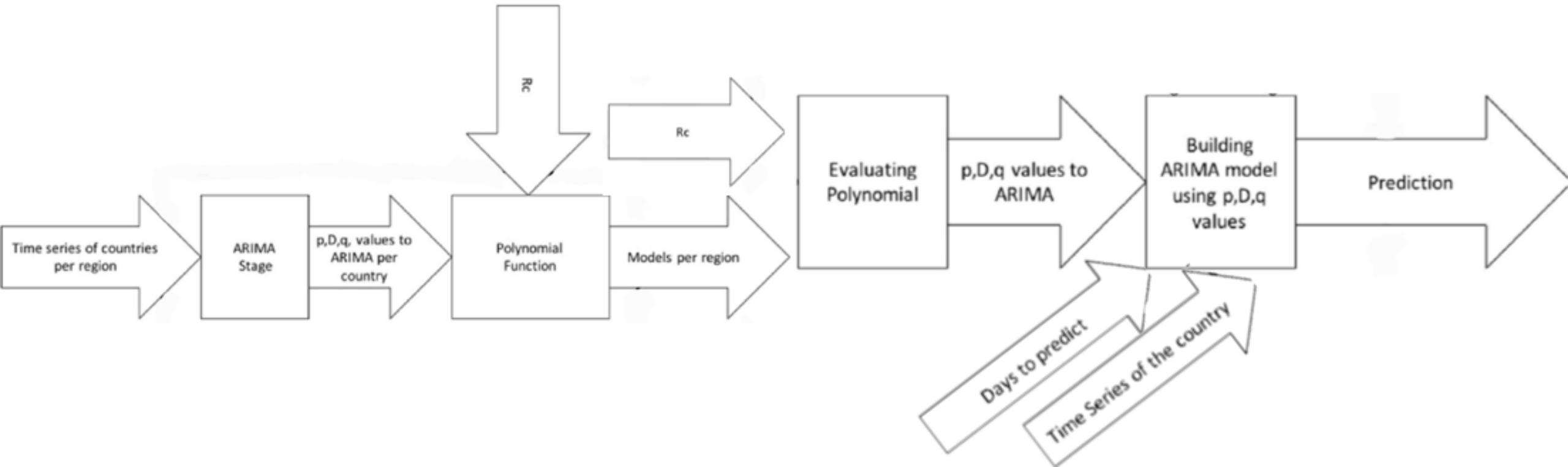
Explore the data on confirmed COVID-19



Explore our data on COVID-19 testing to

<https://ourworldindata.org/>

Methodology



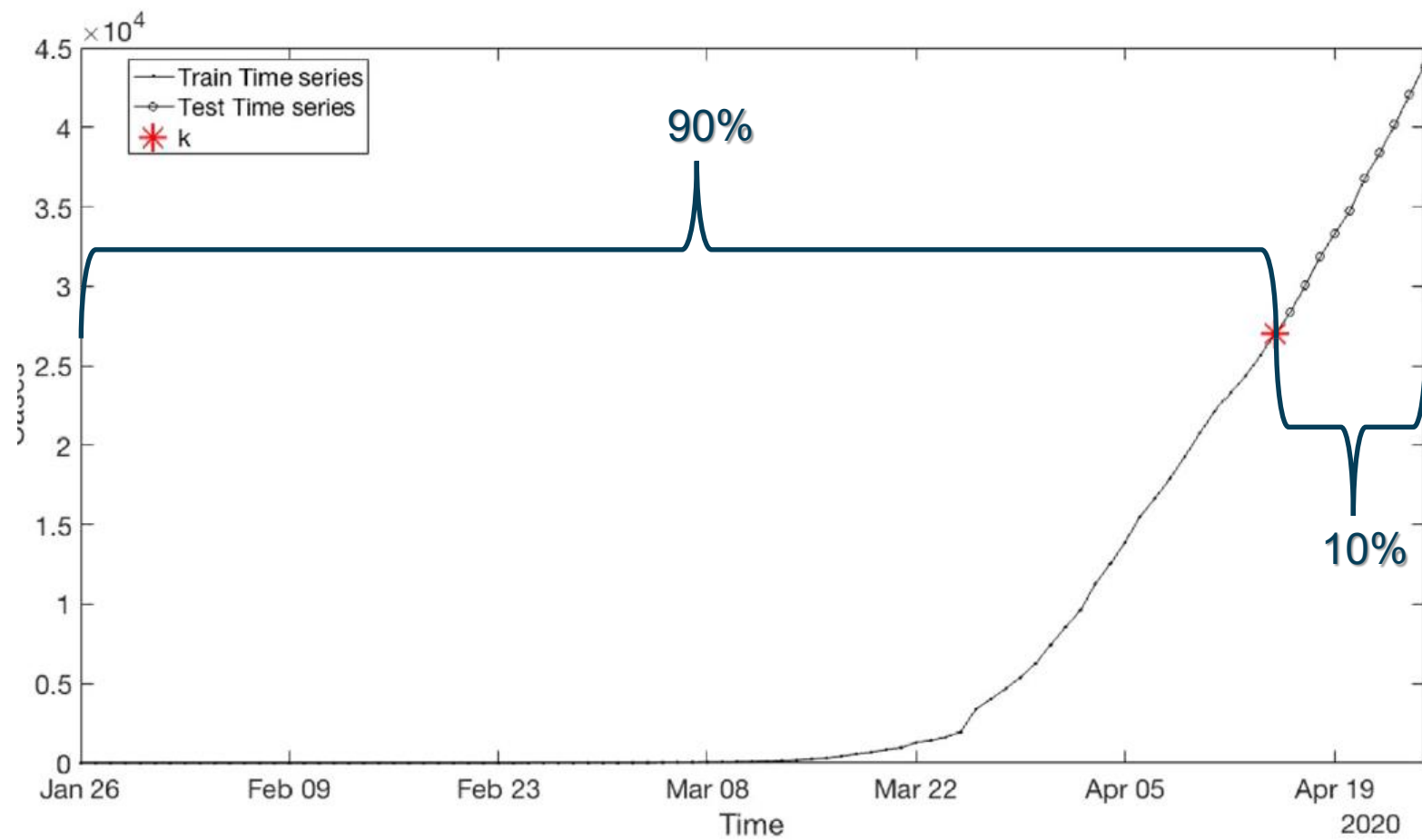


Fig. 4. Canada time series separated into training and testing time series.

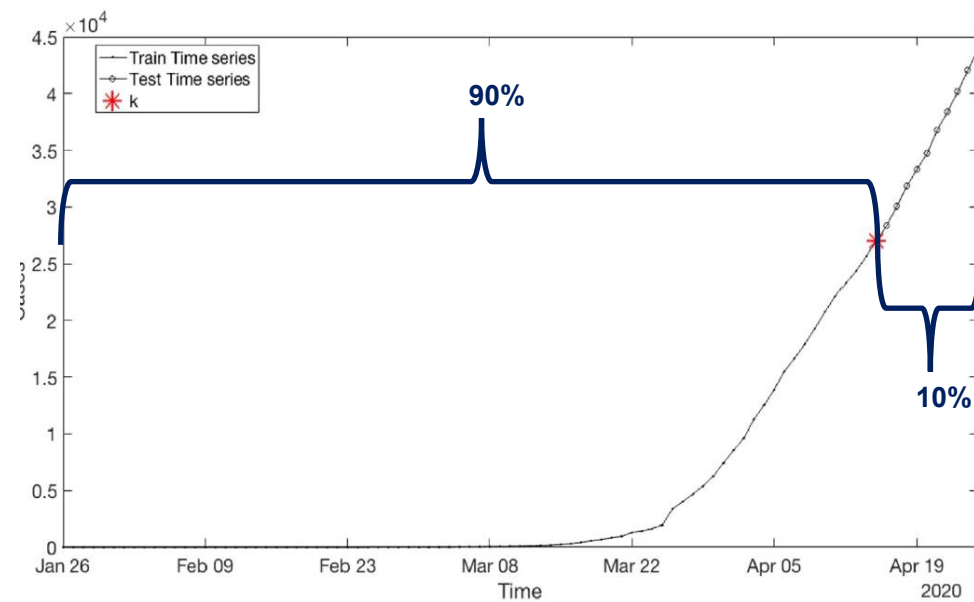
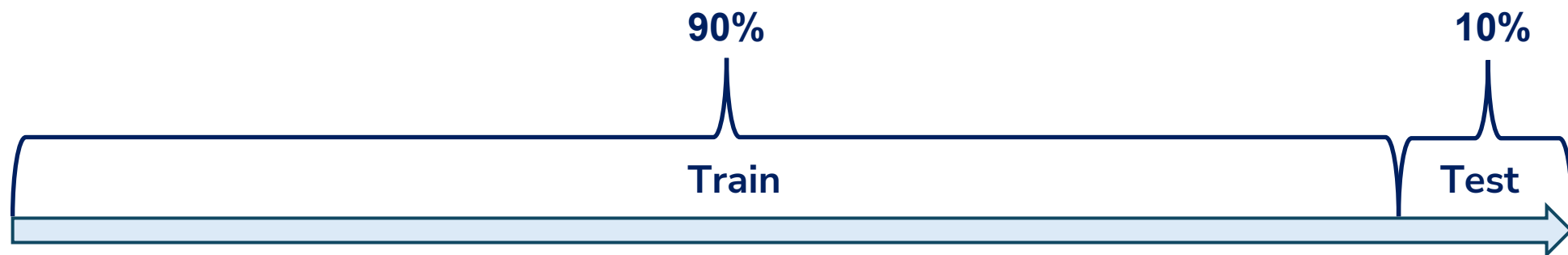


Fig. 4. Canada time series separated into training and testing time series.



1st Case in the country

25.4.2020



$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$



$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

Iterations

```

for px = 0 to 5 do
    for dx = 0 to 5 do
        for qx= 0 to 5 do
  
```

Fit, Predict

```

    Model←ARIMA(px, dx, qx, Train)
    Forecast←Model(f)
  
```

RMSE, p,d,q vectors

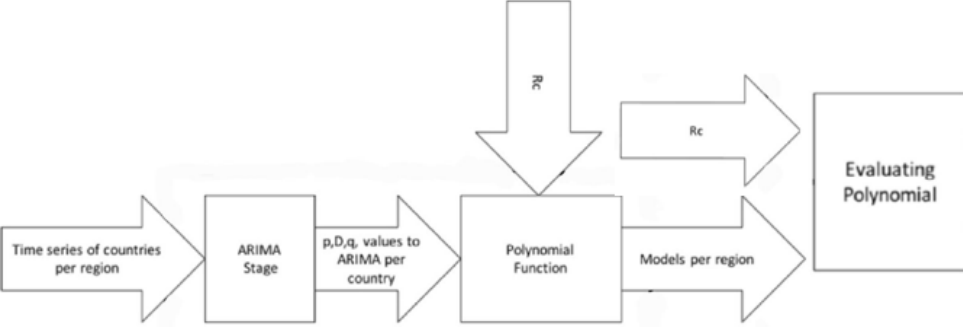
```

    vector(j) ←RMSE(Forecast,Test)
    Pval(j) ←px
    Dval(j) ←dx
    Qval(j) ←qx
  
```

Min RSME and
corresponding p,d,q

```

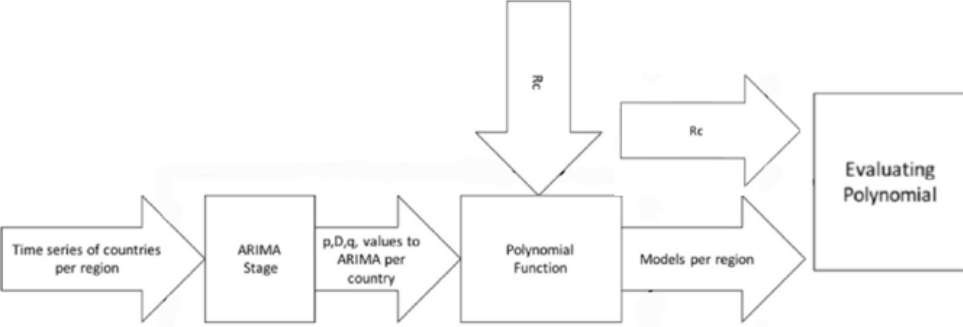
    x←index of min(vector)
    p←Pval(x)
    D←Dval(x)
    q←Qval(x)
  
```



Continental Polynomial

What is RC?

$$\text{RC} = \text{Cases per million people} = \frac{\text{Confirmed cases}}{\text{Population}} \times 1,000,000$$



Continental Polynomial

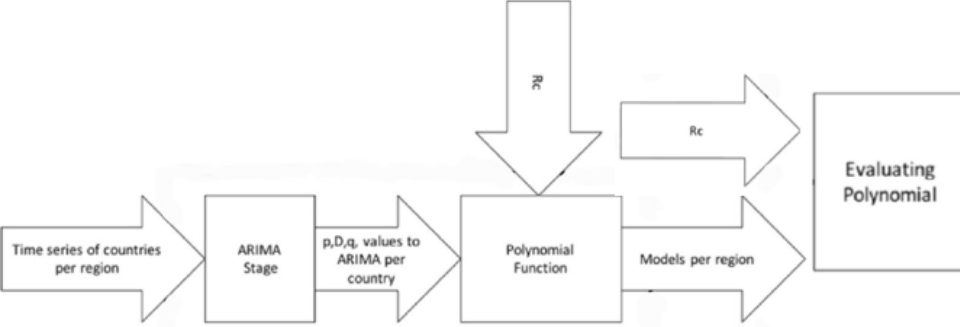
What is RC?

$$RC = \text{Cases per million people} = \frac{\text{Confirmed cases}}{\text{Population}} \times 1,000,000$$

Example:

- Confirmed cases: 10,000
- Population: 5,000,000

$$R_c = \frac{10,000}{5,000,000} * 1,000,000 = 2,000$$



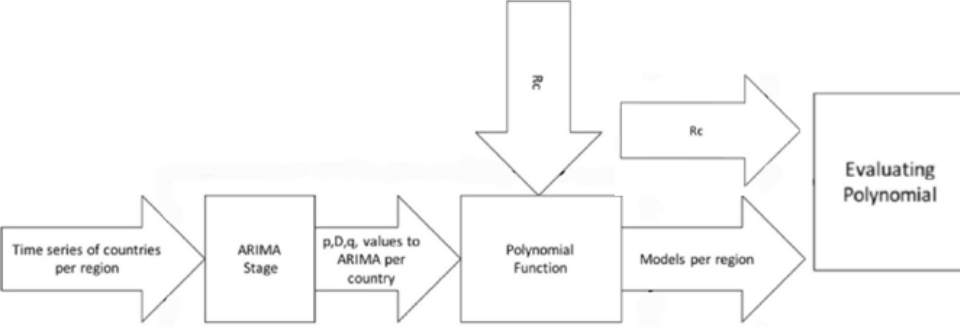
“Population measurement”

$$RC = \frac{\text{Confirmed cases}}{\text{Population}} \times 1,000,000$$

Parameter Vectors

Build 4 ‘parameter vectors’ for each continent

$$V_{p_{NA}} = \begin{pmatrix} p_{C_1} \\ p_{C_2} \\ \vdots \\ p_{C_k} \end{pmatrix}, V_{d_{NA}} = \begin{pmatrix} d_{C_1} \\ d_{C_2} \\ \vdots \\ d_{C_k} \end{pmatrix}, V_{q_{NA}} = \begin{pmatrix} q_{C_1} \\ q_{C_2} \\ \vdots \\ q_{C_k} \end{pmatrix}, V_{R_{cNA}} = \begin{pmatrix} R_{cc_1} \\ R_{cc_2} \\ \vdots \\ R_{cc_k} \end{pmatrix}$$



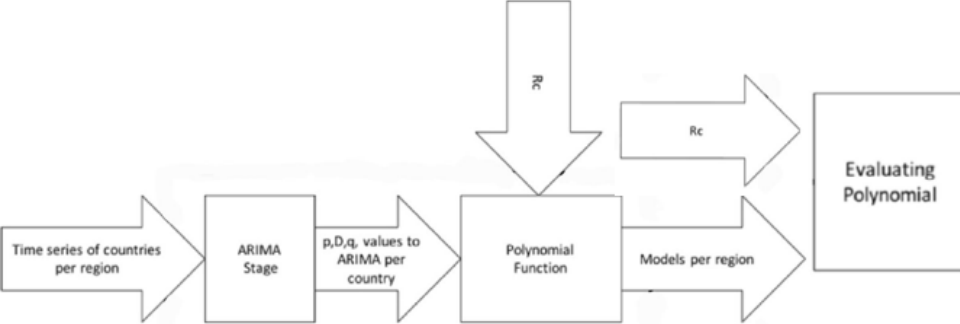
"Population measurement"

$$\boxed{\text{RC}} = \frac{\boxed{\text{Confirmed cases}}}{\boxed{\text{Population}}} \times \boxed{1,000,000}$$

$$V_{p_{NA}} = \begin{pmatrix} p_{C_1} \\ p_{C_2} \\ \vdots \\ p_{C_k} \end{pmatrix}, V_{d_{NA}} = \begin{pmatrix} d_{C_1} \\ d_{C_2} \\ \vdots \\ d_{C_k} \end{pmatrix}, V_{q_{NA}} = \begin{pmatrix} q_{C_1} \\ q_{C_2} \\ \vdots \\ q_{C_k} \end{pmatrix}, V_{R_{cNA}} = \begin{pmatrix} R_{cC_1} \\ R_{cC_2} \\ \vdots \\ R_{cC_k} \end{pmatrix}$$

We want to find a polynomial f such that

$$f(R_{cC_i}) = a_0 + a_1(R_{cC_i}) + a_2(R_{cC_i})^2 + \dots + a_n(R_{cC_i})^n = q_{C_i}$$



"Population measurement"

$$\boxed{\text{RC}} = \frac{\boxed{\text{Confirmed cases}}}{\boxed{\text{Population}}} \times \boxed{1,000,000}$$

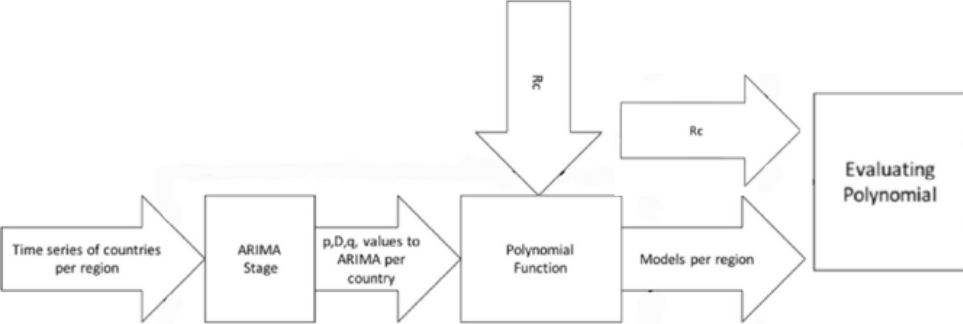
$$V_{p_{NA}} = \begin{pmatrix} p_{C_1} \\ p_{C_2} \\ \vdots \\ p_{C_k} \end{pmatrix}, V_{d_{NA}} = \begin{pmatrix} d_{C_1} \\ d_{C_2} \\ \vdots \\ d_{C_k} \end{pmatrix}, V_{q_{NA}} = \begin{pmatrix} q_{C_1} \\ q_{C_2} \\ \vdots \\ q_{C_k} \end{pmatrix}, V_{R_{cNA}} = \begin{pmatrix} R_{cC_1} \\ R_{cC_2} \\ \vdots \\ R_{cC_k} \end{pmatrix}$$

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$$f'(R_{cC_i}) = a'_0 + a'_1(R_{cC_i}) + a'_2(R_{cC_i})^2 + \dots + a'_n(R_{cC_i})^n = p_{C_i}$$

$$f''(R_{cC_i}) = a''_0 + a''_1(R_{cC_i}) + a''_2(R_{cC_i})^2 + \dots + a''_n(R_{cC_i})^n = d_{C_i}$$



"Population measurement"

RC

$$= \frac{\text{Confirmed cases}}{\text{Population}} \times 1,000,000$$

$$V_{p_{NA}} = \begin{pmatrix} p_{C_1} \\ p_{C_2} \\ \vdots \\ p_{C_k} \end{pmatrix}, V_{d_{NA}} = \begin{pmatrix} d_{C_1} \\ d_{C_2} \\ \vdots \\ d_{C_k} \end{pmatrix}, V_{q_{NA}} = \begin{pmatrix} q_{C_1} \\ q_{C_2} \\ \vdots \\ q_{C_k} \end{pmatrix}, V_{R_{cNA}} = \begin{pmatrix} R_{cC_1} \\ R_{cC_2} \\ \vdots \\ R_{cC_k} \end{pmatrix}$$

$$f(R_{cC_i}) = a_0 + a_1(R_{cC_i}) + a_2(R_{cC_i})^2 + \dots + a_n(R_{cC_i})^n = qC_i$$

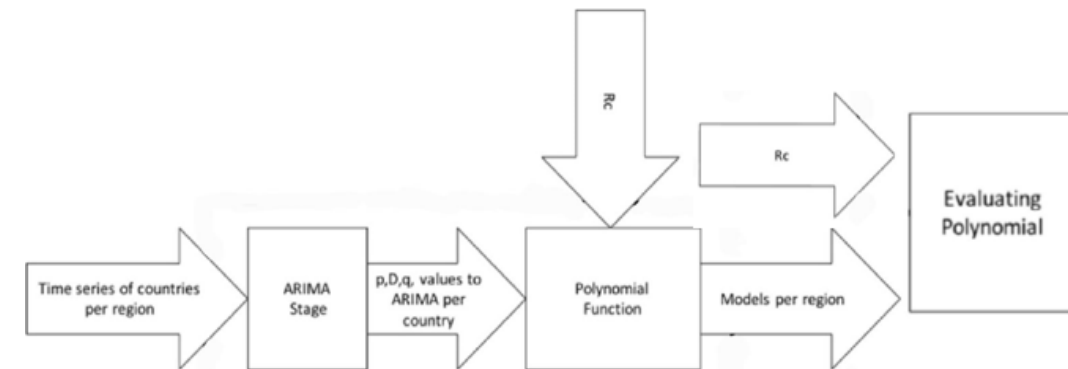
Polynomial extrapolation - Vandermonde matrix

$$\begin{pmatrix} 1 & t_0 & t_0^2 & \dots & t_0^n \\ 1 & t_1 & t_1^2 & \dots & t_1^n \\ \vdots & \dots & \dots & \dots & \vdots \\ 1 & t_m & t_m^2 & \dots & t_m^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} qC_{(1)} \\ qC_{(2)} \\ \vdots \\ qC_{(m)} \end{pmatrix}$$

Polynomial extrapolation Vandermonde matrix

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What about the n?

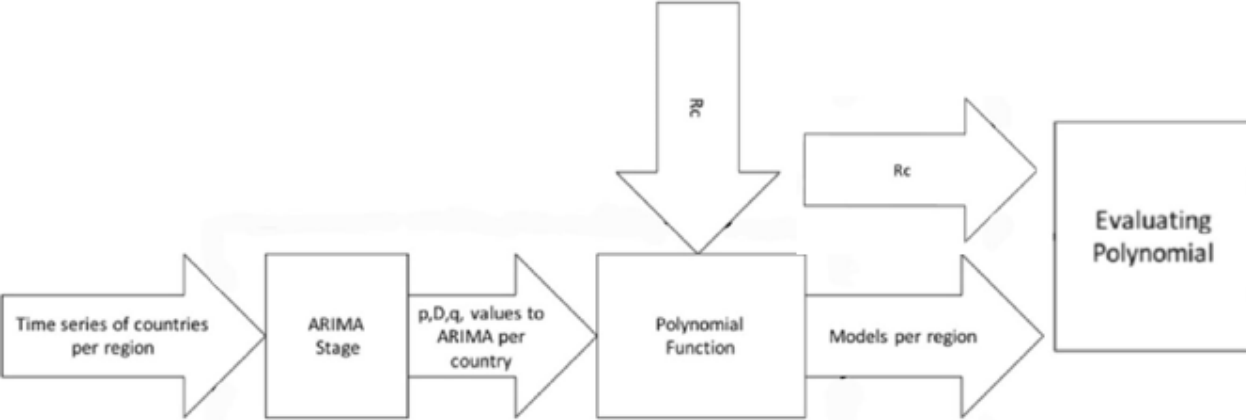


"Population measurement"

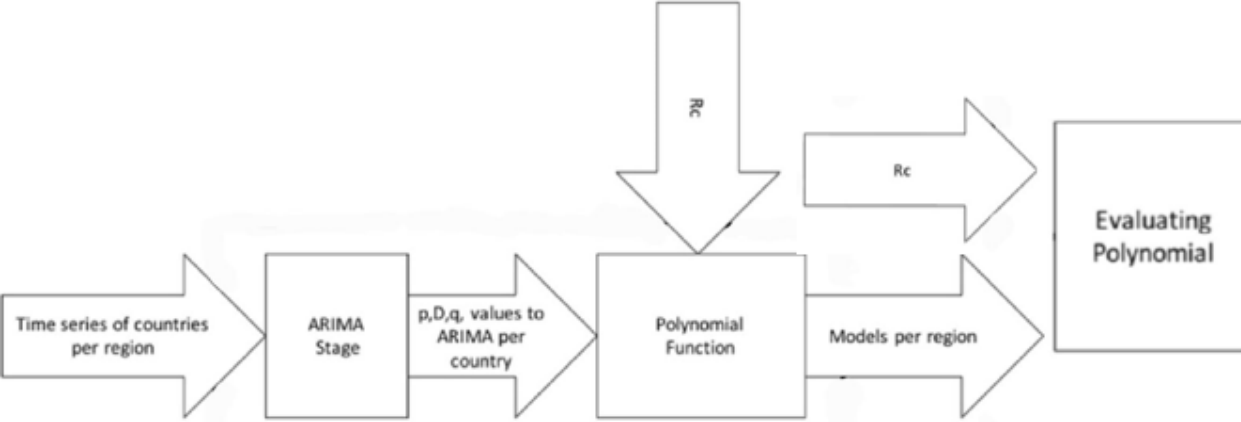
$$\boxed{RC} = \frac{\boxed{\text{Confirmed cases}}}{\boxed{\text{Population}}} \times \boxed{1,000,000}$$

$$V_{pNA} = \begin{pmatrix} pC_1 \\ pC_2 \\ \vdots \\ pC_k \end{pmatrix}, V_{dNA} = \begin{pmatrix} dC_1 \\ dC_2 \\ \vdots \\ dC_k \end{pmatrix}, V_{qNA} = \begin{pmatrix} qC_1 \\ qC_2 \\ \vdots \\ qC_k \end{pmatrix}, V_{RcNA} = \begin{pmatrix} RcC_1 \\ RcC_2 \\ \vdots \\ RcC_k \end{pmatrix}$$

$$f(RcC_i) = a_0 + a_1(RcC_i) + a_2(RcC_i)^2 + \dots + a_n(RcC_i)^n = qC_i$$



$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$



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$$\begin{pmatrix} 1 & t_0 & t_0^2 & \cdots & t_0^n \\ 1 & t_1 & t_1^2 & \cdots & t_1^n \\ \vdots & \cdots & \cdots & \cdots & \vdots \\ 1 & t_m & t_m^2 & \cdots & t_m^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} qC_{(1)} \\ qC_{(2)} \\ \vdots \\ qC_{(m)} \end{pmatrix}$$

Iterations

For p, D, q do

For i = 1 to 100 do

Solve
Vandermonde
matrix

$P_n \leftarrow$ solve the equation 16 using i value as degree of polynomial

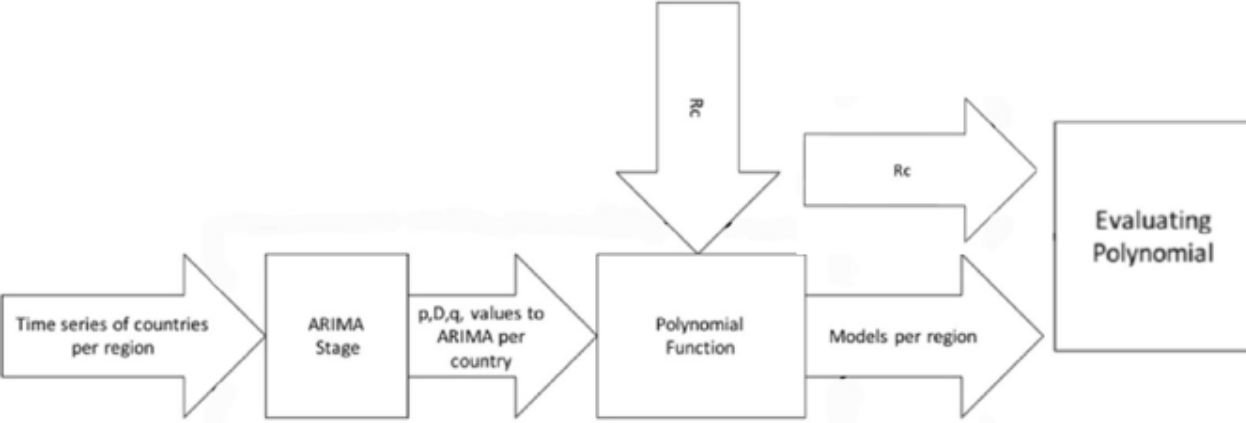
$y \leftarrow$ evaluate the polynomial using P_n and t

$y \leftarrow \text{round}(y)$

$\text{rmse}(i) \leftarrow$ calculate the RMSE between y and d

$\text{polynomial degree} \leftarrow$ index of min(rmse)

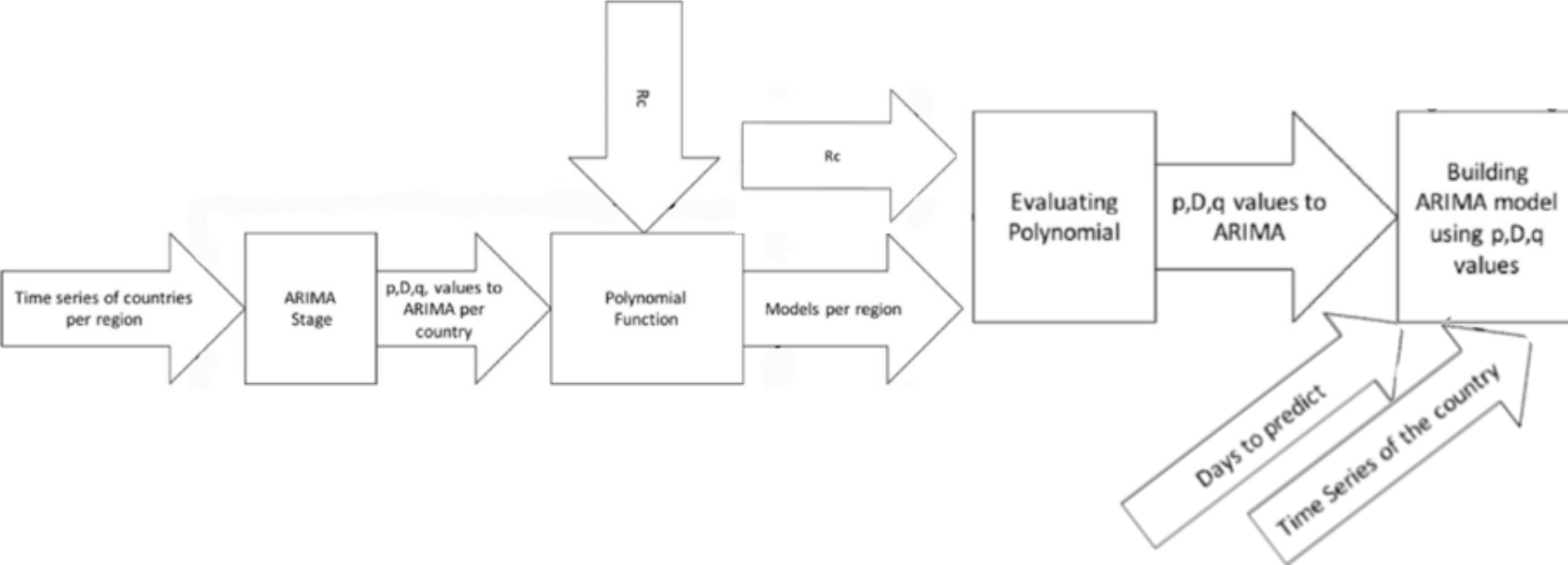
Min RSME and
corresponding degree



$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$$

$$\begin{pmatrix} 1 & t_0 & t_0^2 & \cdots & t_0^n \\ 1 & t_1 & t_1^2 & \cdots & t_1^n \\ \vdots & \cdots & \cdots & \cdots & \vdots \\ 1 & t_m & t_m^2 & \cdots & t_m^n \end{pmatrix} \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} q_{C_{(1)}} \\ q_{C_{(2)}} \\ \vdots \\ q_{C_{(m)}} \end{pmatrix}$$

$$\begin{aligned} p_{NA_d}(t) = & p_1 t^{15} + p_2 t^{14} + p_3 t^{13} + p_4 t^{12} + p_5 t^{11} \\ & + p_6 t^{10} + p_7 t^9 + p_8 t^8 + p_9 t^7 + p_{10} t^6 \\ & + p_{11} t^5 + p_{12} t^4 + p_{13} t^3 + p_{14} t^2 + p_{15} t \\ & + p_{16} \end{aligned}$$



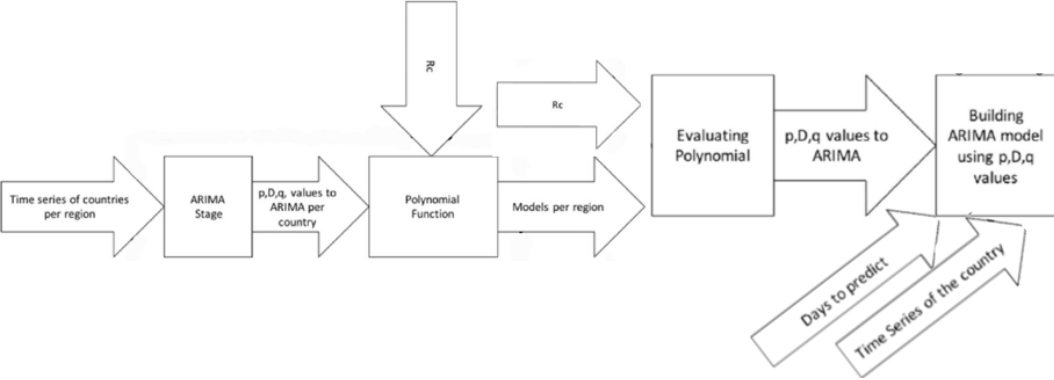
Now we have **Continental Polynomial** for each continent.

For example: for Canada, we use the NA functions

Canada's Rc on
11.5.20

$$\left\{ \begin{array}{l} p_{NAp}(t) = p_1 t^n + p_2 t^{n-1} + p_3 t^{n-2} + \dots + p_n t + p_{n+1}; n = 14 \\ p_{NA d}(t) = p_1 t^n + p_2 t^{n-1} + p_3 t^{n-2} + \dots + p_n t + p_{n+1}; n = 15 \\ p_{NA q}(t) = p_1 t^n + p_2 t^{n-1} + p_3 t^{n-2} + \dots + p_n t + p_{n+1}; n = 47 \end{array} \right.$$

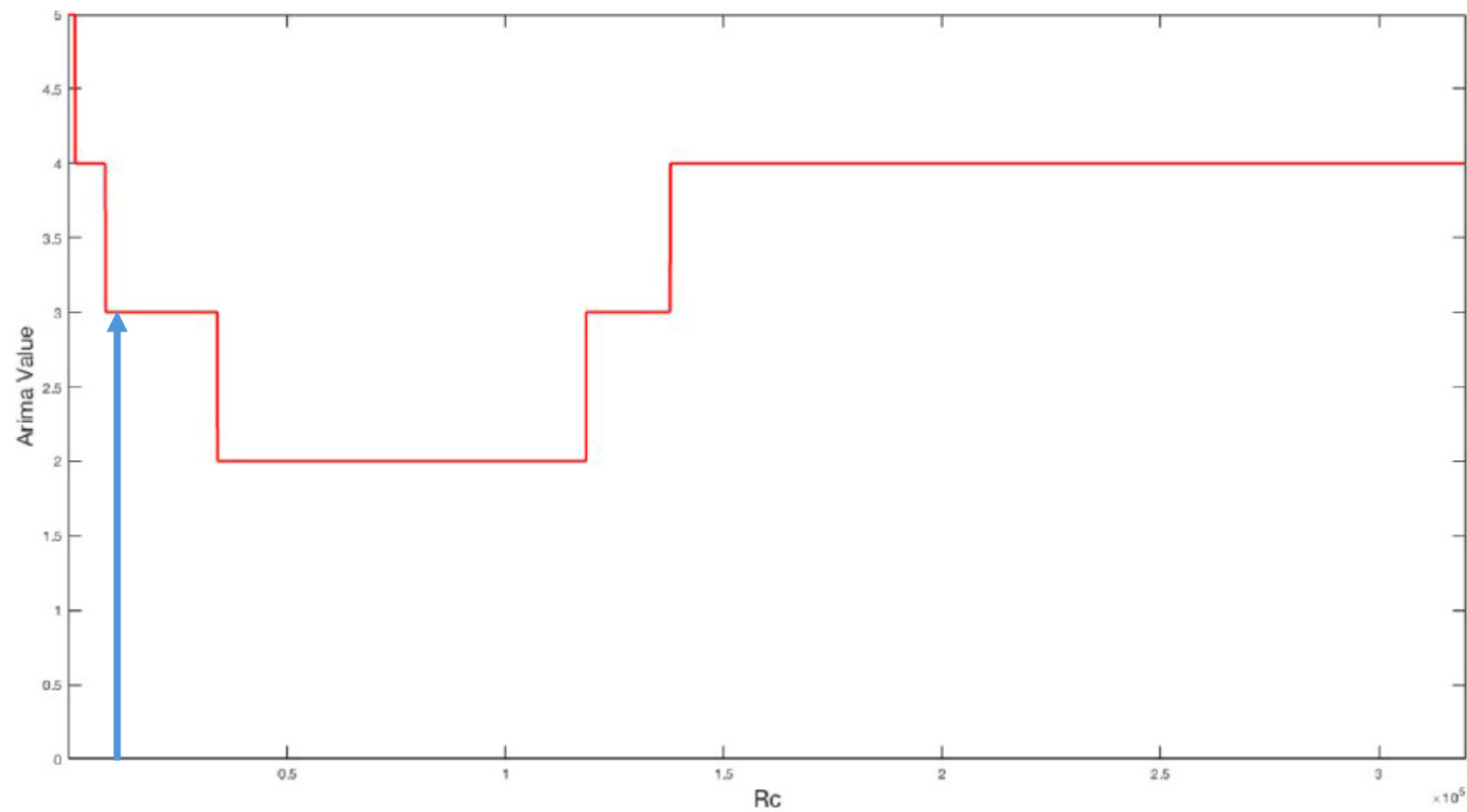
p= 3
d= 2
q= 3



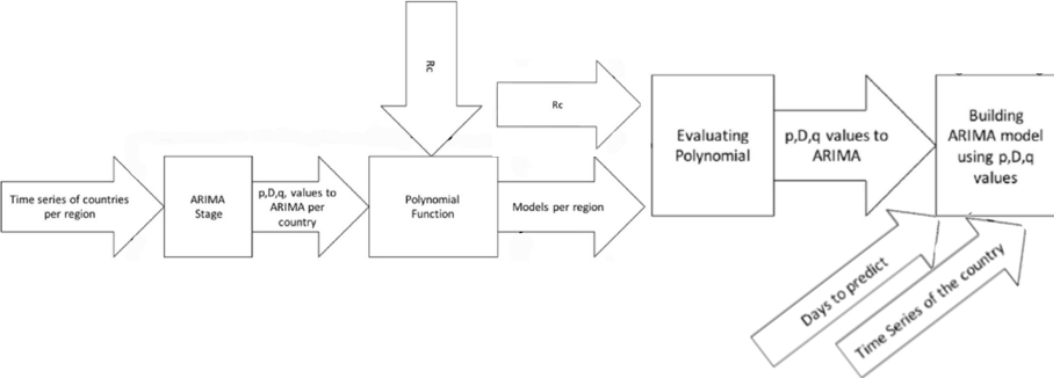
$p = 3$

$d = 2$

$q = 3$



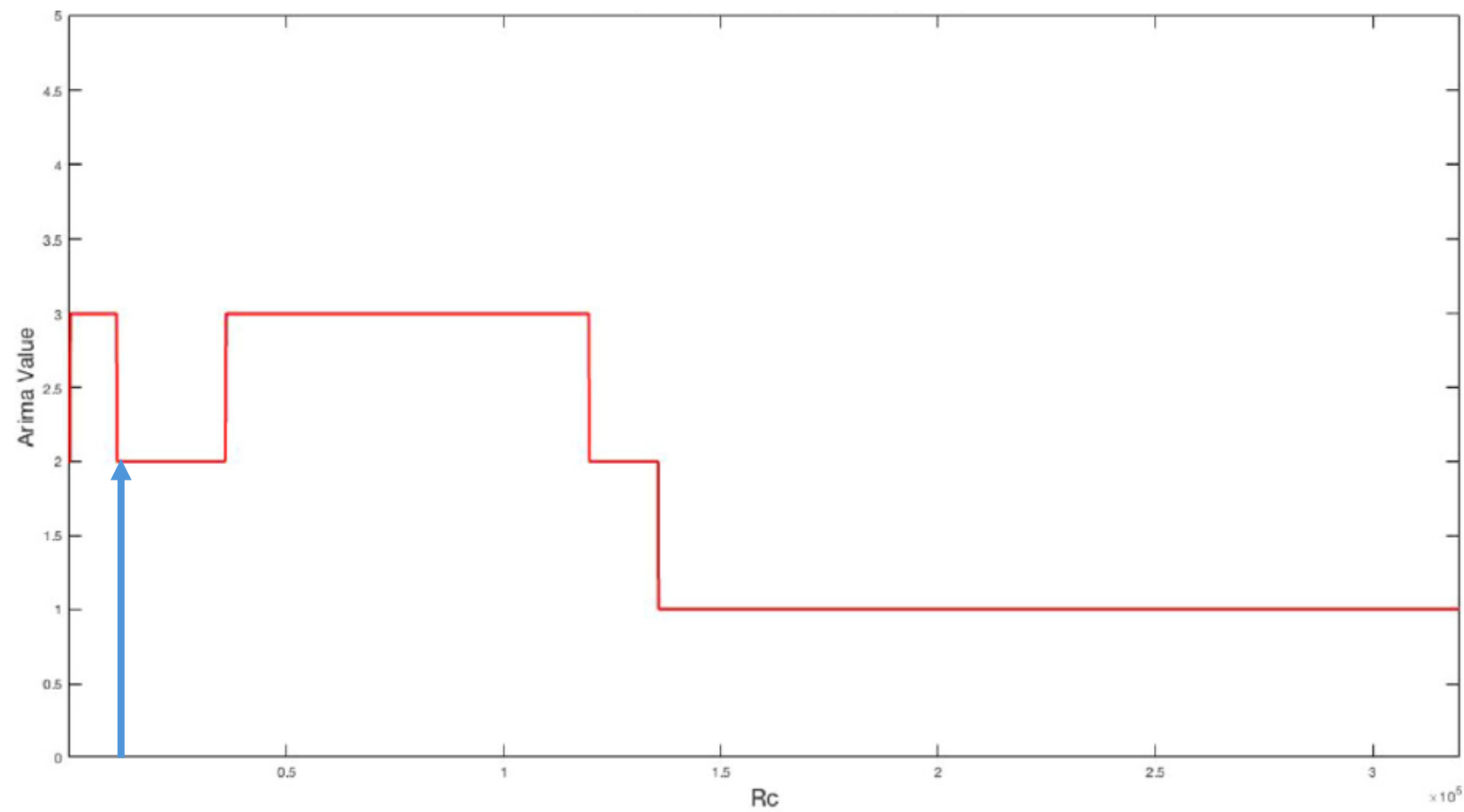
(A) Parameter p .



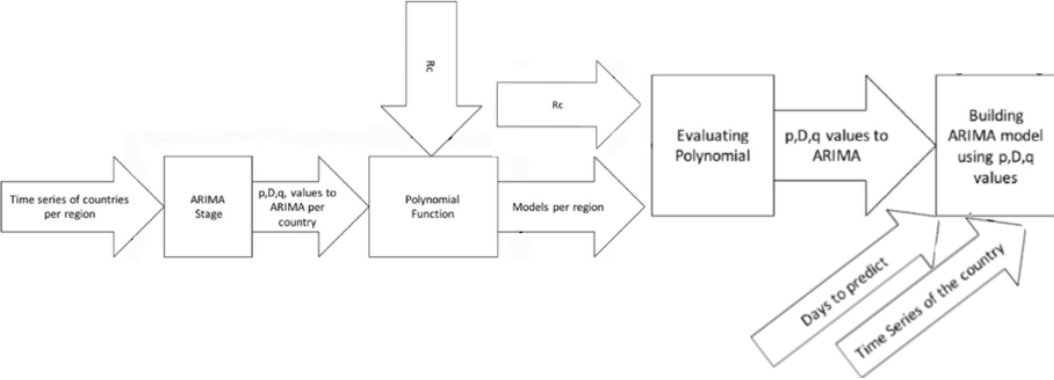
$p = 3$

$d = 2$

$q = 3$



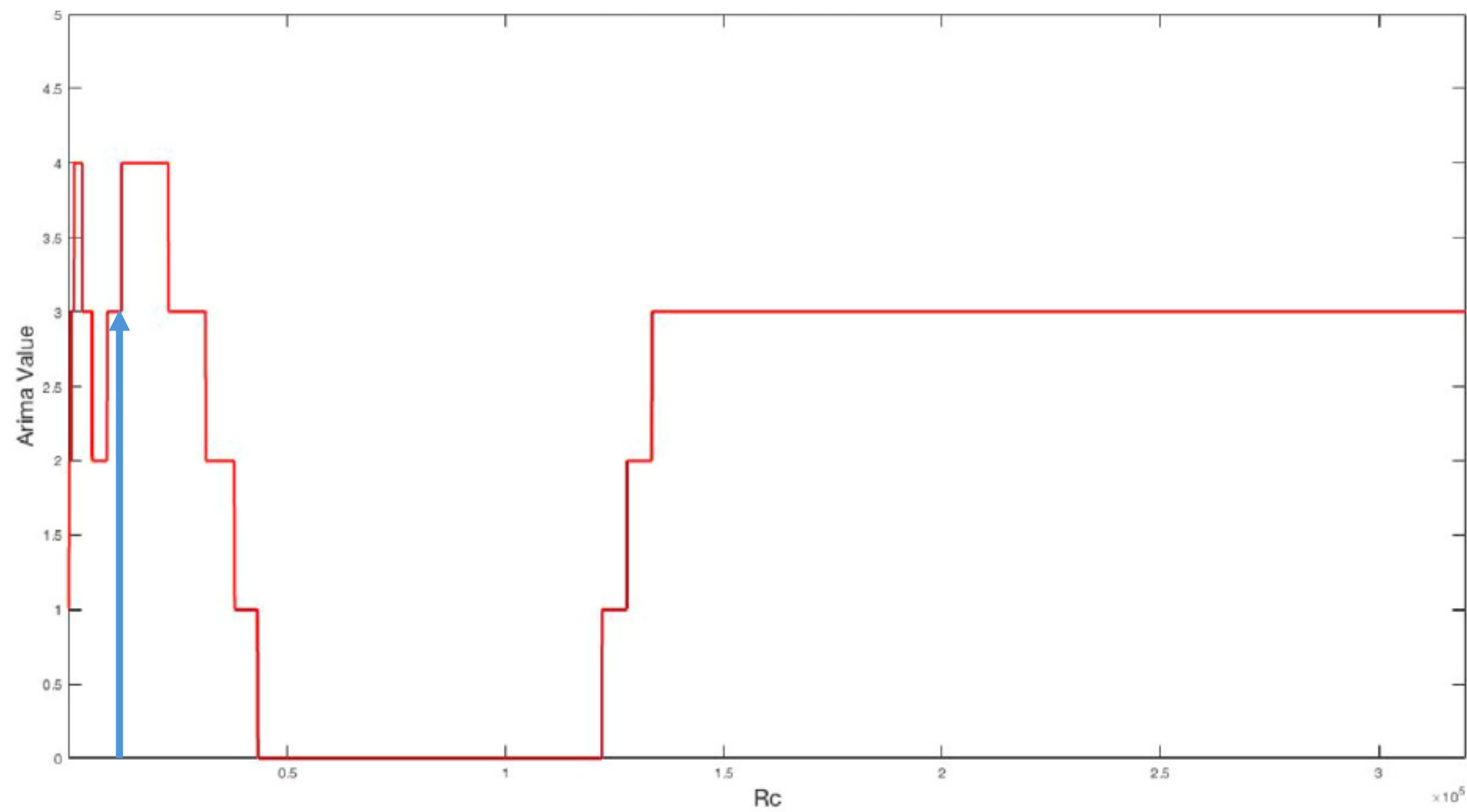
(B) Parameter D .



$p = 3$

$d = 2$

$q = 3$



(C) Parameter q .

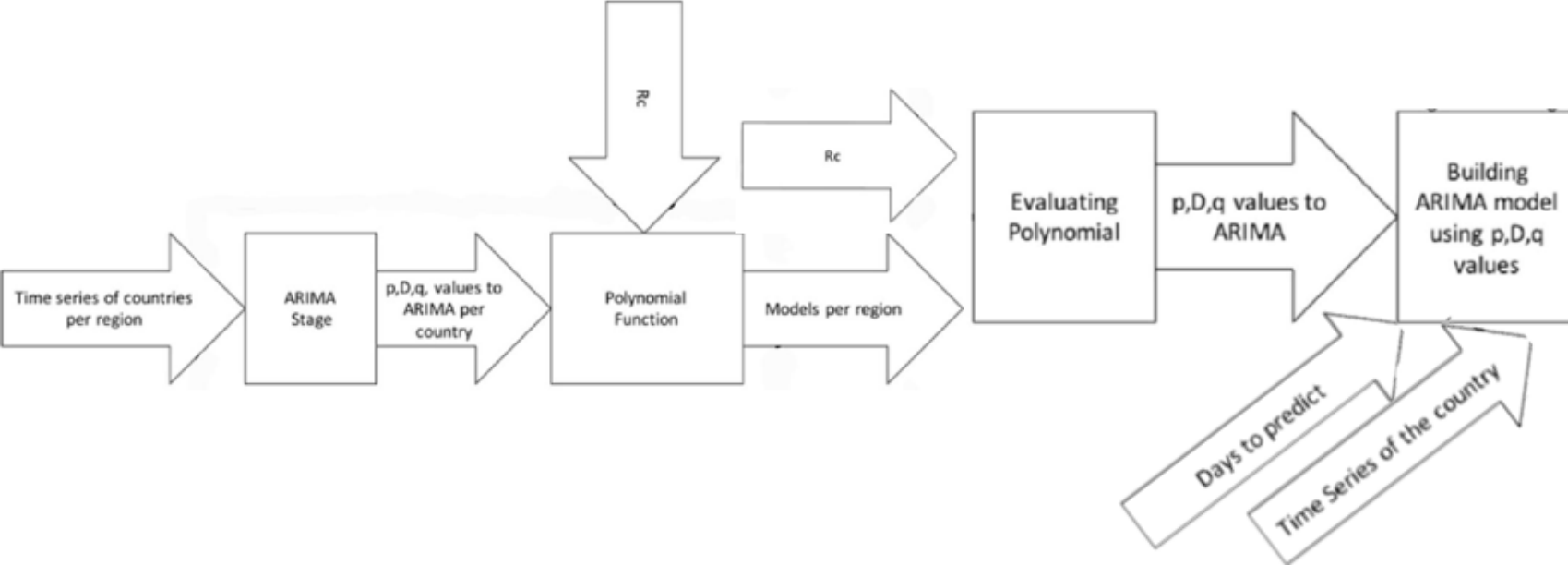
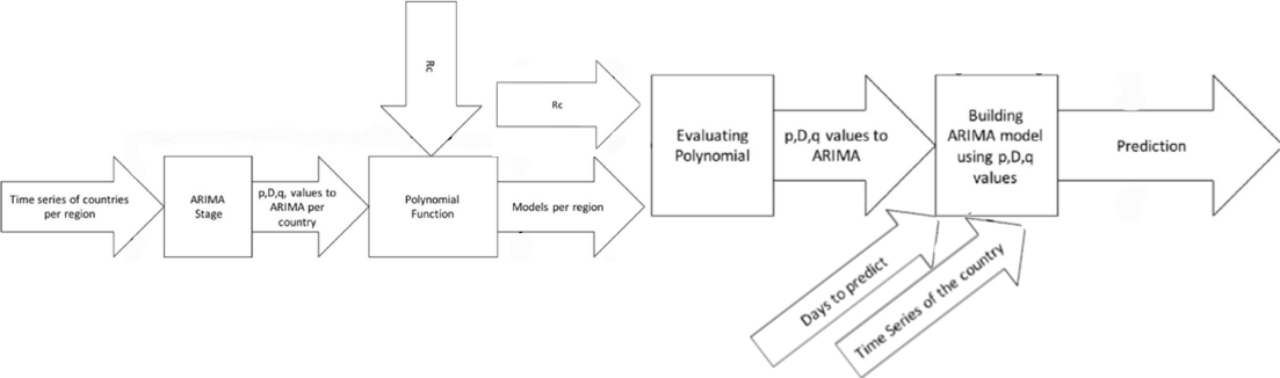
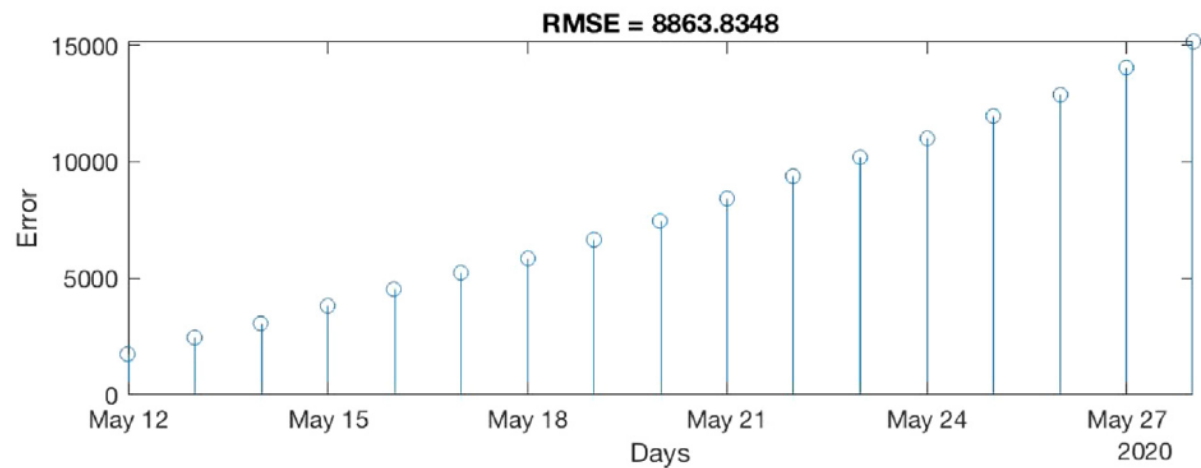
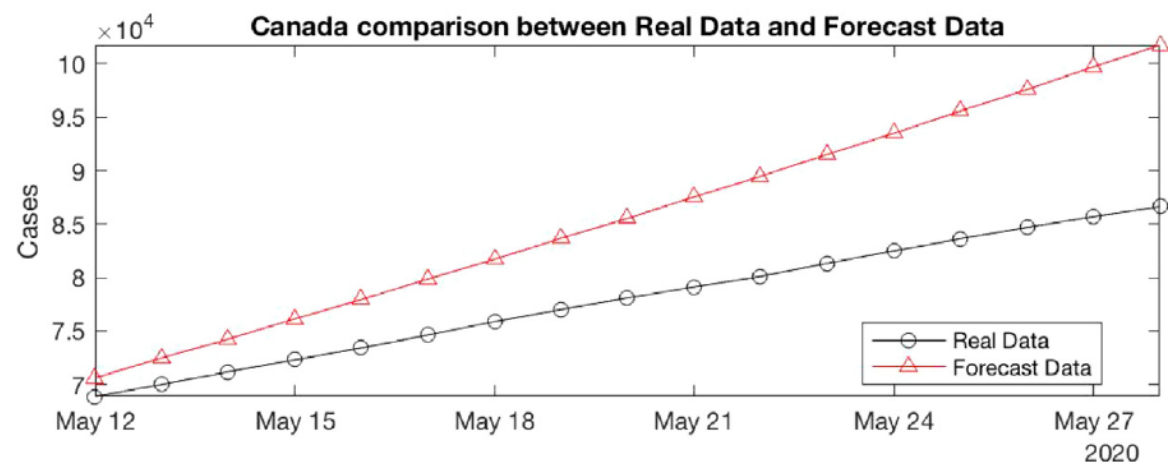
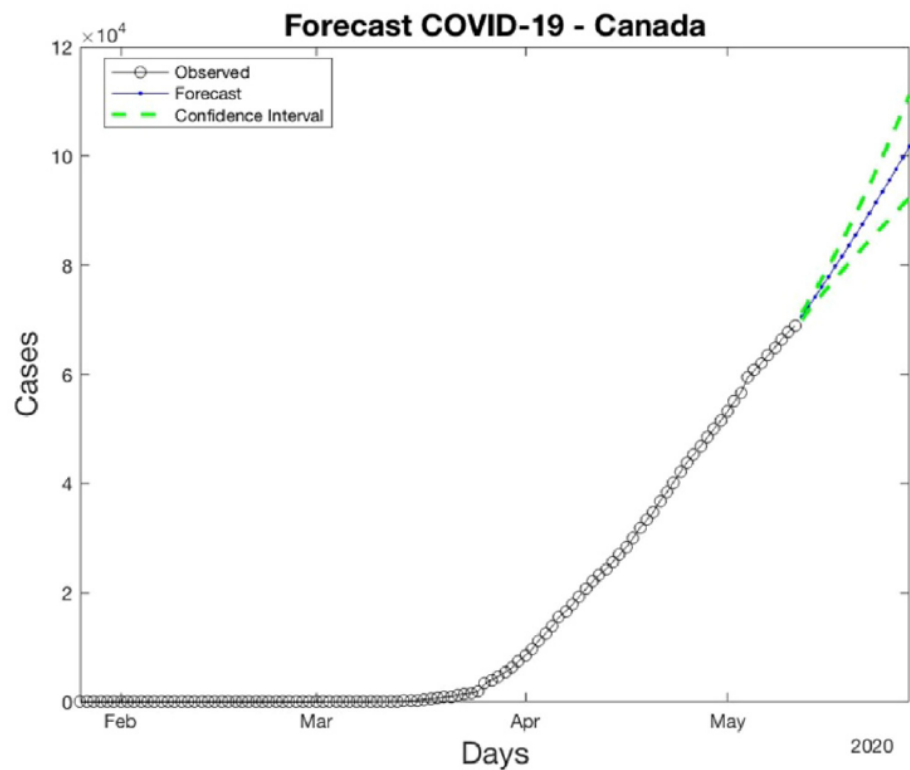


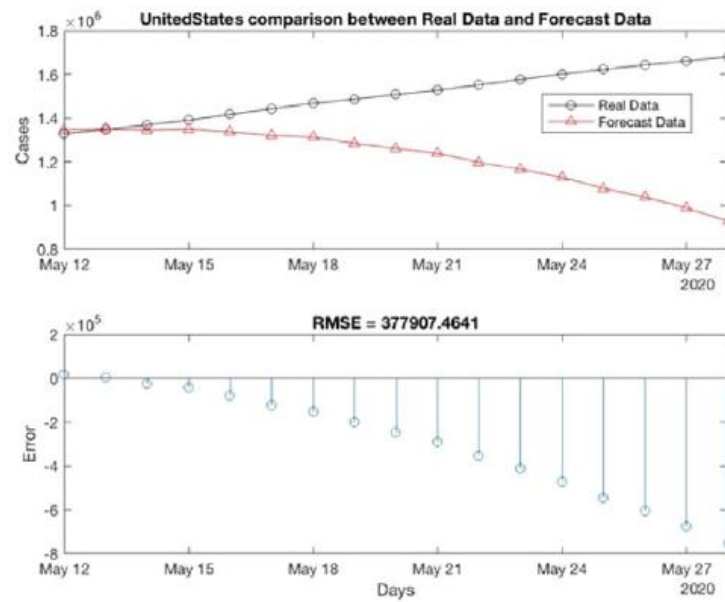
Table 1
Data of North America.

Country	Population per million people (ppMp)	Total confirmed COVID-19 cases per 1M people		ARIMA Parameters			RMSE average 640.61
		April 25	May-11	p	D	q	
Belize	0.390	45.269	45.269	0	2	0	2.830E−07
Canada	37.411	1162.546	1824.167	3	2	3	2.611E+02
Cuba	11.333	113.450	155.916	5	1	3	2.128E+01
Dominican Republic	10.739	529.964	953.825	5	0	3	6.851E+01
Grenada	0.112	133.311	186.635	5	1	3	2.452E−01
Guatemala	17.581	24.001	58.720	2	2	5	1.671E+01
Haiti	11.263	6.314	15.961	3	3	1	2.392E+00
Honduras	9.746	59.669	199.099	4	3	1	1.719E+01
Jamaica	2.948	97.259	169.528	5	3	3	7.446E+00
Mexico	127.576	99.835	271.630	2	3	0	1.614E+02
Nicaragua	6.546	1.660	2.415	5	5	3	3.859E−01
Panama	4.246	1237.146	1957.927	3	1	5	2.141E+01
United States	329.065	2690.383	4017.488	4	1	3	7.750E+03

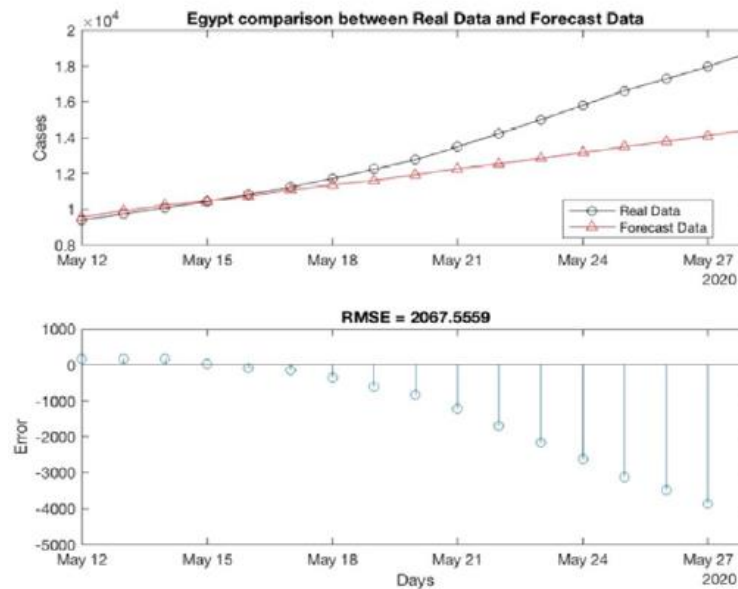


Results





(a.1) United States comparison.



(c.1) Egypt comparison.

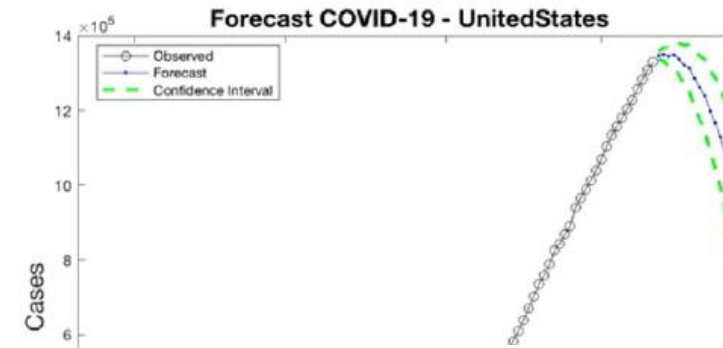
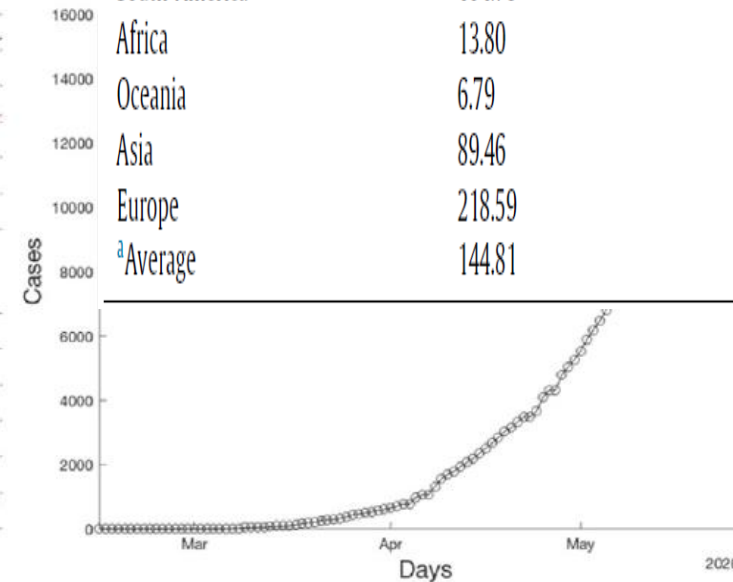


Table 2

Average of RMSE results.

Region	Average of RMSE between original and forecast signal in training stage	Average of RMSE between original and forecast signal from May 12 to May 27
North America	640.61	3.6051e+04
South America	104.78	2.0828e+04
Africa	13.80	1.4913e+03
Oceania	6.79	161.2570
Asia	89.46	3.52964e+03
Europe	218.59	2.88212e+04
Average	144.81	1.2723e+04



(c.2) Egypt Forecast.

Table 2

Average of RMSE results.

Region	Average of RMSE between original and forecast signal in training stage	Average of RMSE between original and forecast signal from May 12 to May 27
North America	640.61	3.6051e+04
South America	104.78	2.0828e+04
Africa	13.80	1.4913e+03
Oceania	6.79	161.2570
Asia	89.46	3.52964e+03
Europe	218.59	2.88212e+04
^a Average	144.81	1.2723e+04

Researchers Conclusions

ARIMA models effectively produce low RMSE, indicating their strong performance in modeling and evaluating COVID-19 spread.

This research demonstrates a method to generalize COVID-19 spread from specific cases, which can be used in future and more complex research.

There exists a correlation between virus spread and various regional variables, such as population size, influencing prediction accuracy (RMSE).

Also, Future research could explore additional factors weather conditions, and cultural or behavioral clusters to enhance prediction accuracy.

The shown model allows for good predictions and with regular updates with the latest data can be continued relevance.



Our conclusions

RMSE Vs
population

Data size and
model
accuracy.

Impact of this
model early
on

Future
models



Summary

- RMSE results are notably low relative to the population sizes in each region, with RMSE values in the thousands versus populations in the millions.
- The article utilized data from the initial outbreak until May 2020, covering only a few months, yet achieved accurate predictions and partially modeled the virus's behavior.
- The researchers successfully developed a model that has some good prediction accuracy despite limited available data, providing decision-makers with a significant tool that noticeably influenced various sectors including public services, commerce, and more.
- The study laid a solid foundation for future research, demonstrating the effectiveness of using simple models in the process.